2024 CpSc Q1

Section: Computer Systems

Topic: Number Representation — Two’s Complement (8-bit)

Question Summary

(a) Convert −25 (denary) to an 8-bit two’s complement binary number.
(b) State the largest positive integer representable in 8-bit two’s complement.

Worked Solution

(a) Converting −25 to 8-bit two’s complement:
1) Write +25 in 8-bit binary: 25 = 16 + 8 + 1 → 00011001.
2) Invert all bits (one’s complement): 11100110.
3) Add 1 to get two’s complement: 11100110 + 00000001 = 11100111.
Therefore, −25 in 8-bit two’s complement is 11100111.

(b) In 8-bit two’s complement, the range is −128 to +127.
Therefore, the largest positive integer representable is 127.

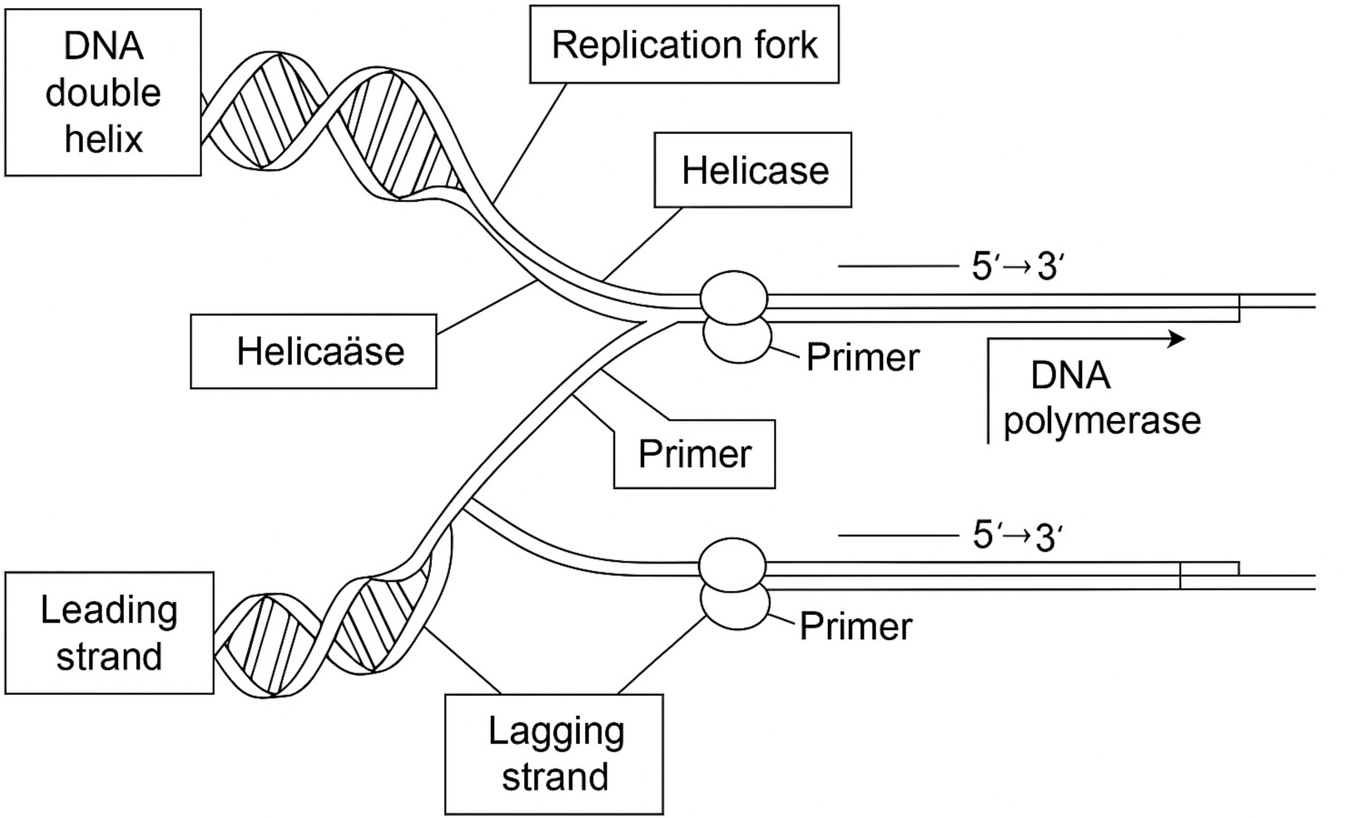
Final Answer

(a) 11100111
(b) 127

Revision Tips

- For −N in two’s complement: write +N in binary, invert bits, add 1.

- Check your result by adding the positive and negative representations — you should get 0 with a discarded carry.
- 8-bit two’s complement range is −128…+127; with n bits it’s $-2^{n-1} \ldots +2^{n-1}-1$.


DNA double helix
Replication fork
Helicase
5′→3′
Helicaäse
Primer
DNA polymerase
Primer
5′→3′
Leading strand
Primer
Lagging strand